

Integration - Set 1

TMUA practice questions by topic. Specification MM7.1-MM7.6. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 Find the area enclosed between the curve $y = x^2 - 4x$ and the x -axis.

- A $-\frac{32}{8^3}$
- B $\frac{3}{8}$
- C $\frac{16}{8}$
- D $\frac{16}{64}$
- E $\frac{32}{64}$
- F $\frac{32}{3}$

- 2 Given that $\int_1^5 f(x) dx = 7$ and $\int_3^5 f(x) dx = 2$, find $\int_1^3 (2f(x) - 1) dx$.

- A 12
- B 16
- C 8
- D 9
- E 10
- F 3

- 3 A curve has $\frac{dy}{dx} = 6x^2 - 4x + 1$ and passes through $(1, 3)$. Find its equation.

- A $y = 12x - 4$
- B $y = 2x^3 - 4x^2 + x + 4$
- C $y = 2x^3 - 2x^2 + x + 2$
- D $y = 2x^3 - 2x^2 + x + 3$
- E $y = 2x^3 - 2x^2 + x$
- F $y = 2x^3 - 2x^2 + 2$

- 4 The trapezium rule with 2 strips of equal width is used to estimate $\int_0^2 x^2 dx$. Which of the following is true?
- A The estimate is $\frac{8}{3}$ and it is exact.
- B The estimate is 3 and it is an overestimate.
- C The estimate is $\frac{5}{2}$ and it is an overestimate.
- D The estimate is $\frac{5}{2}$ and it is an underestimate.
- E The estimate is 3 and it is an underestimate.
- 5 Given that $\int_0^a (2x + 1) dx = 12$ where $a > 0$, find a .
- A 4
- B 12
- C 2
- D 3
- E $\sqrt{12}$
- F 6
- 6 Let $F(x) = \int_2^x (t^2 - 3t) dt$. Find $F'(3)$.
- A 9
- B $\frac{9}{2}$
- C -3
- D $-\frac{7}{6}$
- E 0
- F 3

7 Which of the following statements is/are true?

I $\int_{-2}^2 x^3 dx = 0$

II The area between $y = x^3$ and the x -axis from $x = -2$ to $x = 2$ is 0.

III $\int_{-2}^2 x^2 dx = 2 \int_0^2 x^2 dx$

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

8 Find $\int \left(x + \frac{1}{x} \right)^2 dx$.

A $\frac{1}{x^3} + 2x - \frac{1}{x} + c$

B $\frac{1}{x^3} - \frac{1}{x} + c$

C $\frac{1}{x^3} \left(x + \frac{1}{x} \right)^3 + c$

D $\frac{1}{x^3} + 2x + \frac{1}{x} + c$

E $\frac{3}{x^3} + 2x + c$

F $x^3 + 2x - \frac{1}{x} + c$

9 Find the value of $\int_1^8 x^{-\frac{2}{3}} dx$.

A $\frac{21}{2}$

B 9

C 3

D 6

E 2

F $\frac{3}{2}$

10 $f(x) = ax + b$ satisfies $\int_0^1 f(x) dx = 2$ and $\int_0^2 f(x) dx = 6$. Find $f(1)$.

- A** 4
- B** 0
- C** 5
- D** 1
- E** 3
- F** 2

11 Given that $\int_0^4 f(x) dx = 10$, find $\int_0^4 (f(x) + 3x) dx - \int_4^0 f(x) dx$.

- A** 20
- B** 0
- C** 14
- D** 44
- E** 24
- F** 34

12 The gradient of a curve at (x, y) is $3\sqrt{x}$ and the curve passes through $(4, 20)$. Find the y-intercept of the curve.

- A** 12
- B** 20
- C** 4
- D** 8
- E** -4
- F** 0